

Practice Sheet: Lecture 2,3,4

Problem Set

Total Marks: 85

Instructions:

- Show all calculations clearly.
- Use base-2 logarithm for entropy.
- For decision tree problems, use ID3 unless otherwise stated.
- For regression tree problems, choose the split with minimum SSR.
- For random forest problems, use majority voting for classification.

Problem 1:

[10 Marks]

A smart library wants to predict whether a student will borrow a book.

Student	Has Exam	Likes Reading	Free Time	Borrow Book?
S1	Yes	Yes	High	Yes
S2	Yes	No	Low	No
S3	No	Yes	High	Yes
S4	No	No	High	No
S5	Yes	Yes	Low	Yes
S6	No	Yes	Low	Yes
S7	Yes	No	High	No
S8	No	No	Low	No

- (a) Calculate the entropy of the full dataset. [2]
- (b) Calculate the information gain for **Likes Reading**. [3]
- (c) Calculate the information gain for **Free Time**. [3]
- (d) Which feature should be selected as the root node? Justify mathematically. [2]

Problem 2:**[8 Marks]**

A music app wants to predict whether a user will replay a song.

User	Genre	Duration	Mood	Replay?
U1	Pop	Short	Happy	Yes
U2	Pop	Long	Sad	No
U3	Rock	Short	Happy	Yes
U4	Rock	Long	Sad	No
U5	Pop	Short	Sad	Yes
U6	Rock	Short	Sad	No

(a) Calculate the weighted Gini impurity for splitting by **Duration**. [3]

(b) Calculate the weighted Gini impurity for splitting by **Mood**. [3]

(c) Which feature gives the better split? [2]

Problem 3:**[12 Marks]**

A coaching center wants to predict final exam score using weekly study hours and attendance.

Student	Study Hours	Attendance	Final Score
S1	1	Low	40
S2	2	Low	42
S3	3	High	45
S4	6	Low	78
S5	7	High	82
S6	8	High	85

A regression tree is constructed using the following rule:

Choose the split with the minimum SSR.

Stopping condition:

Stop splitting a node if it contains fewer than 3 samples.

At the root node, consider the following candidate splits:

$$\textit{Study Hours} < 2.5, \quad \textit{Study Hours} < 4.5, \quad \textit{Study Hours} < 6.5$$

and

$$\textit{Attendance} = \textit{Low/High}$$

- (a) For each candidate split, compute the left and right child predictions using the average Final Score. [3]
- (b) Compute the SSR for each candidate split. [4]
- (c) Select the best root split. [1]
- (d) After applying the best root split, determine whether each child node should be split further using the stopping condition. [1]
- (e) For any child node that is allowed to split, calculate the best next split using SSR. [2]
- (f) Draw the final regression tree and write the prediction value in each leaf. [1]

Problem 4:

[8 Marks]

A food delivery app wants to predict order amount using distance and membership status.

Customer	Distance	Member	Order Amount
C1	1	No	100
C2	2	No	110
C3	1	Yes	140
C4	2	Yes	150
C5	7	No	400
C6	8	No	410
C7	7	Yes	460
C8	8	Yes	470

(a) Compute the SSR for the split $Distance < 4.5$. [3]

(b) Within the left branch ($Distance < 4.5$), compute the SSR after splitting by **Member**. [2]

(c) Within the right branch ($Distance \geq 4.5$), compute the SSR after splitting by **Member**. [2]

(d) Write the final four prediction rules using both Distance and Member. [1]

Problem 5:

[7 Marks]

A Random Forest is trained using the following original rows:

$$R_1, R_2, R_3, R_4, R_5, R_6$$

Three bootstrapped datasets are created:

$$B_1 = \{R_1, R_2, R_2, R_4, R_5, R_6\}$$

$$B_2 = \{R_1, R_3, R_3, R_3, R_4, R_6\}$$

$$B_3 = \{R_2, R_4, R_5, R_5, R_6, R_6\}$$

For a new query, the trees trained on these datasets produce:

$$Tree_1 = Yes, \quad Tree_2 = No, \quad Tree_3 = Yes$$

- (a) Identify which rows are repeated in each bootstrapped dataset. [2]
- (b) Identify one row that is absent from each bootstrapped dataset. [2]
- (c) Use majority voting to find the final Random Forest prediction. [1]
- (d) If $Tree_2$ did not use the feature **Humidity**, is that allowed in Random Forest? Answer mathematically/algorithmically. [2]

Problem 6:

[7 Marks]

A Random Forest with 10 trees is used to estimate a missing numerical value for Example 4. The proximity values between Example 4 and other examples are:

$$Proximity(Ex4, Ex1) = 0.2$$

$$Proximity(Ex4, Ex2) = 0.3$$

$$Proximity(Ex4, Ex3) = 0.5$$

Known income values are:

$$Income(Ex1) = 20, \quad Income(Ex2) = 40, \quad Income(Ex3) = 80$$

- (a) Explain what $Proximity(Ex4, Ex3) = 0.5$ means in terms of 10 trees. [2]
- (b) Normalize the proximity values. [2]

- (c) Estimate the missing income value of Example 4 using proximity-weighted average. [3]

Problem 7:

[15 Marks]

A sports club wants to predict whether a student will join the evening practice session.

Student	Weather	Energy	Homework	Join Practice?
S1	Sunny	High	No	Yes
S2	Sunny	Low	Yes	No
S3	Rainy	High	No	Yes
S4	Rainy	Low	Yes	No
S5	Sunny	High	Yes	Yes
S6	Rainy	Low	No	No

A Random Forest with three trees is constructed. For each tree, a bootstrapped dataset and a random subset of features are given.

Use Gini impurity to construct each tree.

$$Gini = 1 - \sum_{i=1}^k p_i^2$$

For each split, choose the feature with the minimum weighted Gini impurity.

Tree 1

Bootstrapped Dataset 1:

Student	Weather	Energy	Homework	Join Practice?
S1	Sunny	High	No	Yes
S2	Sunny	Low	Yes	No
S2	Sunny	Low	Yes	No
S3	Rainy	High	No	Yes
S5	Sunny	High	Yes	Yes
S6	Rainy	Low	No	No

Randomly selected features for Tree 1:

Energy, Homework

- (a) Calculate the weighted Gini impurity for splitting by **Energy**. [2]

(b) Calculate the weighted Gini impurity for splitting by **Homework**. [2]

(c) Select the root feature for Tree 1. [1]

Tree 2

Bootstrapped Dataset 2:

Student	Weather	Energy	Homework	Join Practice?
S1	Sunny	High	No	Yes
S3	Rainy	High	No	Yes
S3	Rainy	High	No	Yes
S4	Rainy	Low	Yes	No
S4	Rainy	Low	Yes	No
S6	Rainy	Low	No	No

Randomly selected features for Tree 2:

Weather, Homework

(d) Calculate the weighted Gini impurity for splitting by **Weather**. [2]

(e) Calculate the weighted Gini impurity for splitting by **Homework**. [2]

(f) Select the root feature for Tree 2. [1]

Tree 3

Bootstrapped Dataset 3:

Student	Weather	Energy	Homework	Join Practice?
S2	Sunny	Low	Yes	No
S2	Sunny	Low	Yes	No
S3	Rainy	High	No	Yes
S5	Sunny	High	Yes	Yes
S5	Sunny	High	Yes	Yes
S6	Rainy	Low	No	No

Randomly selected features for Tree 3:

Weather, Energy

(g) Calculate the weighted Gini impurity for splitting by **Weather**. [2]

(h) Calculate the weighted Gini impurity for splitting by **Energy**. [2]

(i) Select the root feature for Tree 3. [1]

Final Random Forest Prediction

A new student has the following information:

Weather = Sunny, Energy = High, Homework = No

(j) Use the three constructed trees to predict the class for the new student. [2]

(k) Find the final Random Forest prediction using majority voting. [1]

Section A: Theory Questions

[20 Marks]

Answer the following questions briefly and precisely.

A1. Classification Tree

[5 Marks]

(a) What is the main difference between a classification tree and a regression tree? [1]

(b) In a classification tree, what does a leaf node represent? [1]

- (c) Why do we prefer a split that produces pure child nodes? [1]
- (d) In ID3, which attribute is selected as the splitting attribute? [1]
- (e) Why is weighted entropy used after splitting instead of simply averaging child entropies? [1]

A2. Regression Tree**[5 Marks]**

- (a) What does a regression tree predict at a leaf node? [1]
- (b) Why does a regression tree use average target value as the leaf prediction? [1]
- (c) What is a residual in a regression tree? [1]
- (d) Why do we square residuals when calculating SSR? [1]
- (e) In regression tree construction, why is the split with minimum SSR selected? [1]

A3. Random Forest**[6 Marks]**

- (a) Why is Random Forest called an ensemble model? [1]
- (b) What does bootstrapping mean in Random Forest? [1]
- (c) Why can the same row appear more than once in a bootstrapped dataset? [1]

- (d) Why does Random Forest use random feature subsets? [1]
- (e) For classification, how does Random Forest make the final prediction? [1]
- (f) State one advantage of Random Forest over a single decision tree. [1]

A4. Missing Values and Proximity in Random Forest [4 Marks]

- (a) Why can proximity not be calculated directly when a row has missing feature values? [1]
- (b) What is the purpose of making an initial guess for missing values? [1]
- (c) What does it mean if two examples have proximity 0.8 in a Random Forest with 10 trees? [1]
- (d) How is proximity used to estimate a missing numerical value? [1]

End of Practice Sheet